EEG-VAE : Variationnal autoencoder for Electroencephalography signal encoding

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Outline

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   - Preliminaries
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State of the Art in EEG Self-Supervised Learning

Two families of models

- *Transformer-based*: EEGPT, CBraMod, REVE, LUNA, CSBrain
- *Mamba-based*: EEGMamba, FEMBA, LuMamba, BAM-U-Net

All share the same pretraining objective: reconstruction in raw signal space

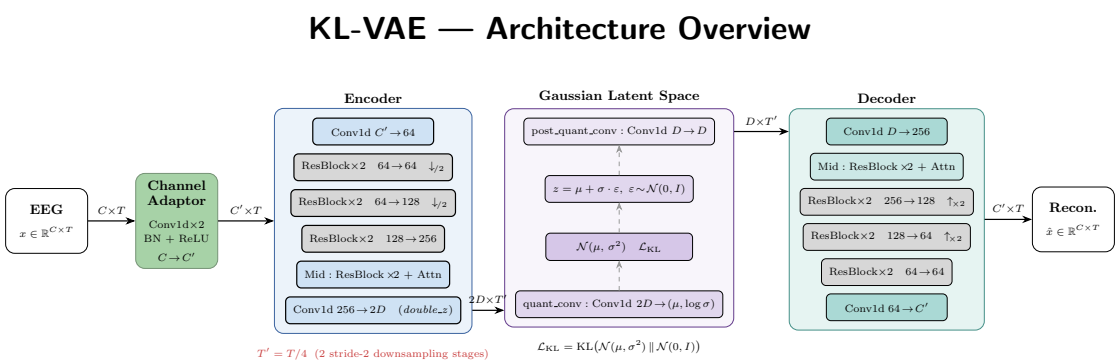
- ✗ Mostly based on Masked Autoencoding - Model capacity wasted on low-level noise & physiological artefacts
- ✗ Patch-based masking ⇒ local pattern matching, not semantic structure
- ✗ No latent-space self-supervised objective

EEG-RQVAE — Our contributions

1. **RVQ-VAE**: produces discrete, hierarchical token sequences — usable for generative & discriminative tasks
2. **2-stage pipeline**: denoising reconstruction (Stage 1) → latent JEPA adaptation (Stage 2), decoupling signal fidelity from semantic structure
3. **SSM analysis**: bidirectional Mamba-2 vs. Transformer encoder
4. **Fair benchmark**: linear probing on 6 (-¿ 12) BCI datasets under identical conditions

Wang et al. 2024 (EEGPT); Wang et al. 2025a (CBraMod); Ouahidi et al. 2026 (REVE); Döner et al. 2025 (LUNA); Zhou et al. 2026 (CSBrain); Wang et al. 2025b (EEGMamba); Tegon et al. 2025 (FEMBA); Broustail et al. 2026 (LuMamba).

KL-VAE: Continuous Latent Space



KL-VAE: Architecture Details and Limitations

Architecture

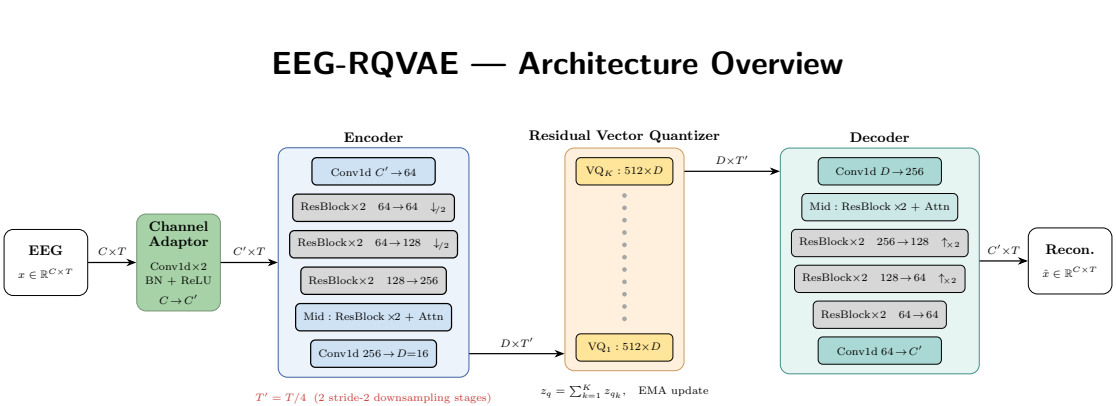
- **Channel adaptor**: Conv1d $\times 2$, BN + ReLU — projects $C \to C' = 64$ without touching time axis
- **Hierarchical encoder**: 3 resolution levels, $d_\ell \in \{64, 128, 256\}$, stride-2 $\Rightarrow T' = T/4$
- **Bottleneck**: 2 ResBlocks + Mid-block, then Conv1d $\to 2D$ (*double_z*)
- **Gaussian latent**: $z = \mu + \sigma \cdot \varepsilon,\ \varepsilon \sim \mathcal{N}(0, I)$, $D = 16$
- **Decoder**: symmetric, $T' \to T$ via nearest-neighbour upsampling

Limitations $\Rightarrow$ motivation for RVQ

- ✗ **Continuous latent**: not tokenizable — cannot feed an autoregressive model
- ✗ **Gaussian prior smooths HF content**: KL pushes z toward $\mathcal{N}(0, I)$, suppressing high-frequency EEG patterns (sleep spindles, γ oscillations)
- ✓ **RVQ** (inspired by EnCodec [1], RQ-VAE [2]): N cascaded codebooks $\Rightarrow$ effective vocabulary K^N with compact per-stage lookup

[1] Défossez et al., *High Fidelity Neural Audio Compression (EnCodec)*, TMLR 2023. [2] Lee et al., *Autoregressive Image Generation using Residual Quantization (RQ-VAE)*, CVPR 2022.

EEG-RQVAE: Discrete Latent Space



EEG-RQVAE: Architecture Details and Advantages

Same backbone as KL-VAE, different bottleneck

- Encoder output $z_e \in \mathbb{R}^{D \times T'}$, $D = 16$, $T' = T/4$
- **RVQ** — $N = 8$ cascaded codebooks, each $K = 512$ entries of dim D:

$$q^{(n)} = \arg\min_k \left\| z^{(n-1)} - e_k^{(n)} \right\|_2, \quad z^{(n)} = z^{(n-1)} - e_{q^{(n)}}^{(n)}$$

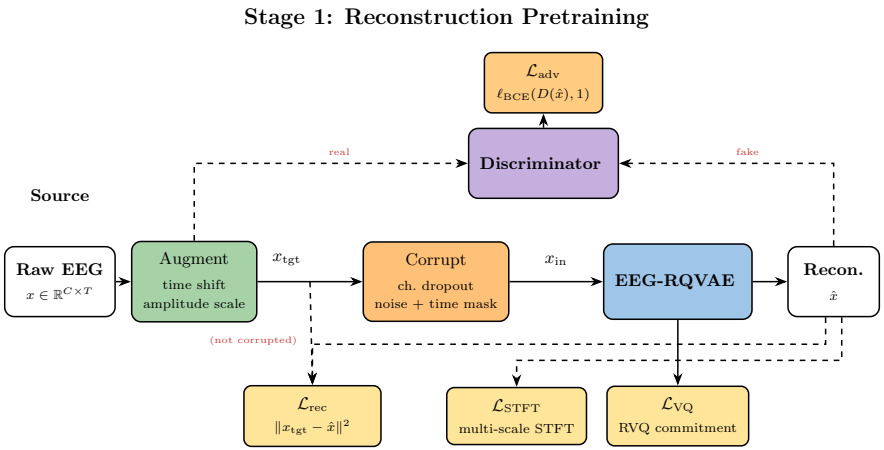
$$z_q = \sum_{n=1}^{N} e_{q^{(n)}}^{(n)}, \qquad \text{effective vocab: } K^N = 512^8$$

- Codebooks updated via **EMA** (decay 0.99) — stable, no auxiliary encoder loss
- Gradients through bottleneck via **straight-through estimator**

Advantages over KL-VAE

- ✓ **Discrete tokens**: N indices per time step — directly usable for generative autoregressive models
- ✓ **No Gaussian prior**: preserves high-frequency EEG content (sleep spindles, γ oscillations)
- ✓ **Hierarchical**: each codebook refines the residual of the previous — coarse-to-fine representation
- ✓ **Scalable**: K^N vocabulary with only $N \times K$ codebook entries stored

Stage 1: Adversarial Denoising Reconstruction



Objective: train encoder–RVQ–decoder to reconstruct a *clean* EEG from a *corrupted* input, while building a compact discrete latent space.

Stage 1: Double-Perturbation Strategy

Why double perturbation?

- A single perturbation lets the model trivially invert the known transform — it learns *nothing* structural
- Two **independent** perturbations force invariance to acquisition-level artefacts (electrode impedance, movement noise)

Step 1 — Augment $\rightarrow$ x_{tgt} *(clean target)*

- Circular time shift $\leq 10\%$ of window
- Amplitude scaling $\sim \mathcal{U}(0.8, 1.2)$
- Channel dropout (up to 2 channels zeroed)

Step 2 — Corrupt $\rightarrow$ x_{in} *(model input, independently)*

- Additional channel dropout
- Gaussian noise ($\sigma = 0.01$)
- Contiguous time masking 5–10%

The decoder is trained to restore x_{tgt} from x_{in} — the model cannot simply copy the input

Stage 1: Training Losses

1. $\mathcal{L}_{\text{rec}} = \text{SmoothL1}(\hat{x},\, x_{\text{tgt}})$ — waveform-level
2. $\mathcal{L}_{\text{spec}} = \dfrac{1}{|\mathcal{N}|} \displaystyle\sum_{n \in \mathcal{N}} \mathcal{L}_{L1}\big(|\text{STFT}_n(\hat{x})|,\, |\text{STFT}_n(x_{\text{tgt}})|\big)$
   $\mathcal{N} = \{64, 128, 256, 512\}$ — covers δ to γ (0.32–2.56 s)
3. $\mathcal{L}_{\text{commit}} = \|z_e - \text{sg}[z_q]\|_2^2$ — encoder $\rightarrow$ codebook
4. $\mathcal{L}_{\text{adv}} = -\mathbb{E}[D(\hat{x})]$ — patch discriminator
   activated from epoch 5; enforces local HF realism

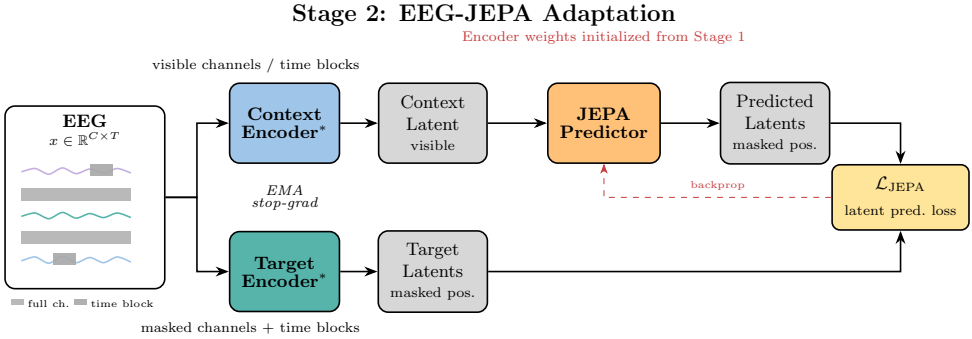
Total Stage 1 objective

$$\mathcal{L}_{\text{Stage 1}} = \mathcal{L}_{\text{rec}} + 0.1\,\mathcal{L}_{\text{spec}} + \mathcal{L}_{\text{commit}} + 0.01\,\mathcal{L}_{\text{ad}}$$

After Stage 1:

- Encoder + RVQ produce physiologically faithful discrete tokens
- **Decoder frozen** — only encoder kept for Stage 2
- Latent space compact but not yet *semantically* structured

Stage 2: Latent JEPA Adaptation



Objective: refine the encoder toward *semantic predictability* in latent space — without any signal-level reconstruction pressure.

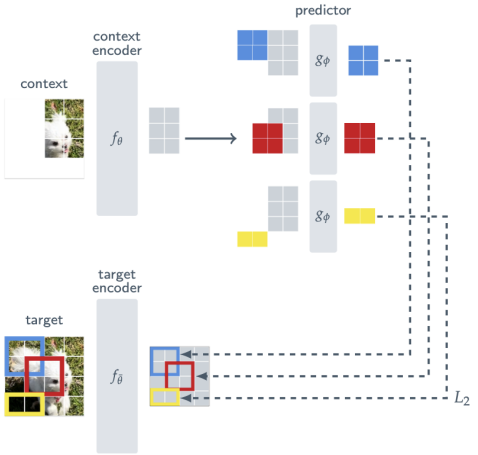
JEPA learning motivation

Masked Autoencoders reconstruct pixels/patches

- ✗ Model capacity wasted on low-level details
- ✗ No semantic structure in the objective
- ✗ Easy shortcuts: predict texture, not meaning

JEPA learns in representation space

- ✓ Predict the **latent representation** of target patches from context
- ✓ Avoids pixel-level reconstruction — focuses on semantics
- ✓ Richer, more abstract features with the same architecture



I-JEPA: Image-based Joint-Embedding Predictive Architecture (Assran et al., 2023)

Stage 2: Model, Masking and Objective

Three components (encoder weights from Stage 1)

- **Context encoder** f_θ: processes visible channels/blocks
- **Target encoder** $f_{\bar{\theta}}$: EMA copy — never updated by gradient
  $\bar{\theta} \leftarrow \tau\bar{\theta} + (1-\tau)\theta, \quad$ cosine warmup $\tau : 1.0 \rightarrow 0.999$
- **Narrow predictor** g_ϕ: compact Transformer (4 blocks, $d_{\text{pred}} = 128$, 4 heads) — less capacity than encoder to prevent trivial solutions

Latent masking on $z_q \in \mathbb{R}^{D \times T_z}$, $T_z = T/4$

- Blocks of $B = 25$ tokens (≈ 500 ms); 25% selected as target $\mathcal{T}$
- Target positions replaced by learnable mask token $m \sim \mathcal{N}(0, 0.02^2)$

JEPA objective

$$\mathcal{L}_{\text{Stage 2}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{s}_t - \text{sg}[s_t]\|_2^2$$

where $\hat{s}_\mathcal{T} = g_\phi(f_\theta(\tilde{z}))$ and $s_\mathcal{T} = f_{\bar{\theta}}(z_q)$

Key insight

- ✓ Loss computed **entirely in latent space** — model never penalised for ignoring HF signal details
- ✓ Biases learning toward **structured temporal dependencies** underlying cognitive dynamics
- ✓ Mirrors MedVAE [1]: decoupling reconstruction from semantic structure improves downstream discriminability

[1] Varma et al., *MedVAE*, 2025.

Datasets

BCI Task	Dataset	Ch.	Dur. (s)	Classes	Subjects	Samples
Pretraining	TUH EEG Corpus (TUEG)	29	30	—	1 385	303 686
Abnormal EEG	TUH Abnormal EEG	26	30	2	1 164	68 486
Sleep staging	Sleep-EDF Expanded (50%)	30	30	5	39	96 362
Emotion	SEED-V	58	1	5	16	115 001
Emotion	FACED	30	10	9	123	10 332
Motor imagery	PhysioNet EEG MMI	64	4	4	109	9 838
Motor imagery	BCI Competition IV-2a	22	4	4	9	4 696
Motor imagery	SHU-MI	32	4	2	25	11 988
Seizure	Siena Scalp EEG	37	30	2	14	16 926
Seizure	CHB-MIT Scalp EEG	23	10	1	24	353 872
Imagined speech	BCI Competition 2020 T3	60	5	5	30	5 250
Imagined speech	KaraOne	58	5	11	14	1 913
Imagined speech	Chisco	62	4	39	5	32 403

Pretraining Hours : 2531 h

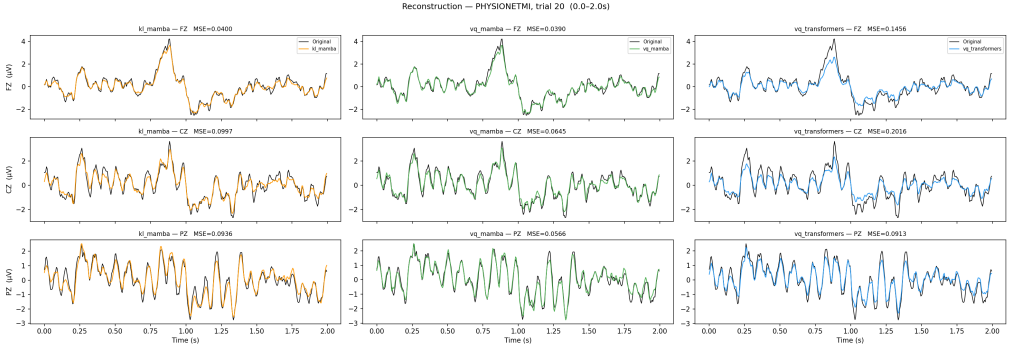
Signal Preprocessing Pipeline

Dataset-agnostic pipeline — same steps applied identically to all 13 datasets

1. **Load** with MNE — resample to **200** Hz
2. **Filter**: band-pass 0.5–75 Hz + notch 60 Hz/50 Hz (IIR)
3. **Channel mapping**: nearest-neighbour projection to fixed $C' = 19$ montage — channels with no source within threshold r zeroed
4. **Per-channel z-score** over each window — flat or non-finite channels neutralised via finite-value mask
5. **Windowing**: fixed-length windows (duration per dataset) — trailing window anchored to segment end, **no zero-padding**
6. **LMDB storage**: window + subject ID, sampling rate, channel names, label

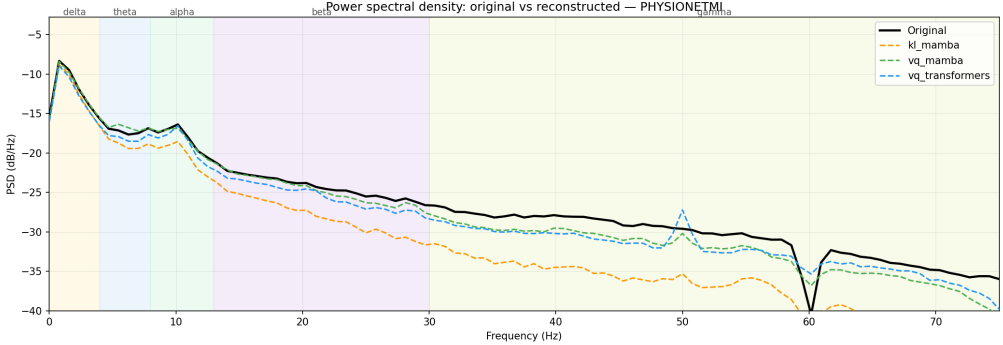
All downstream signals share **identical formatting and sampling rate** as pretraining — no dataset-specific preprocessing

Stage 1: Reconstruction Quality



- **RVQ models** track sharp transients more tightly (lower MSE) than KL-Mamba
- KL-Mamba over-smooths peak amplitudes — Gaussian prior averages fast oscillations

Stage 1: Power Spectral Density

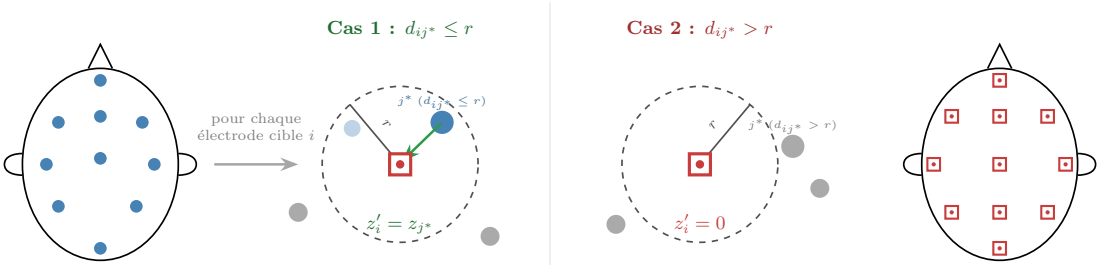


- **VQ models** closely track the original spectrum up to the γ band
- KL-Mamba systematically under-represents high-frequency power due to Gaussian smoothing

Downstream Results — Linear Probing (BACC ↑)

Model	BCI2A	PhysioNetMI	FACED	SEED-V	TUAB	Sleep-EDF
EEGPT	0.347	0.573	0.413	0.269	0.741	0.625
CBraMod	0.439	0.472	0.332	0.278	0.732	0.457
EEGMamba	0.408	**0.578**	**0.471**	0.256	0.732	0.584
REVE	—	—	—	—	—	—
CSBrain	—	—	—	—	—	—
LUNA	—	—	—	—	—	—
LaBraM	—	—	—	—	—	—
BAM-U-Net	—	—	—	—	—	—
KL-Mamba	0.456	0.536	0.306	0.287	—	—
RVQ-Transformer	0.469	0.528	0.262	0.280	—	—
JEPA	0.443	0.538	0.315	0.283	—	—
RVQ-Mamba	**0.484**	0.519	0.325	**0.289**	**0.767**	**0.727**

Ablation: Channel Mapping Strategies



$$j^*(i) = \arg\min_j \; d\big(e_j^{\text{src}},\, e_i^{\text{tgt}}\big) \qquad z'_i = \begin{cases} z_{j^*(i)} & \text{si } d_{ij^*} \leq r \\ 0 & \text{sinon} \end{cases}$$

Ablation: Channel Mapping Results (RVQ-Mamba)

	Raw (native channels)			Mapped (19-ch montage)		
Dataset	BACC	κ	MF1	BACC	κ	MF1
BCI2A	0.483	0.309	0.469	**0.495**	**0.326**	**0.475**
PhysioNetMI	0.565	0.420	0.564	**0.567**	**0.422**	**0.570**
FACED	0.346	0.264	0.342	**0.369**	**0.289**	**0.366**
Sleep-EDF†	**0.727**	**0.592**	**0.560**	0.200	0.000	0.093

† Sleep-EDF has only 2 derivations — mapping zero-fills 17/19 inputs, encoder receives an almost empty montage.

Limitations

Pretraining & scale

- Single corpus (TUEG), 10% of files (1 385 subjects) — too small to claim a foundation model
- Corpus scale and site diversity are primary drivers of transfer

Codebook stability

- Higher residual stages risk under-utilization — effective vocabulary below $K^N = 512^8$
- Sensitive to latent dimension D and EMA decay

Loss interactions

- Stage 1 combines 4 terms with fixed weights — relative contribution uncharacterized

Stage 2 benefit is task-dependent

- Latent JEPA doubles pretraining cost but no uniform gain — trails RVQ-Mamba on BCI2A, FACED, SEED-V
- No independent metric to assess Stage 2's structural contribution

Sequence directionality

- Checkpoints use a *causal* scan — future context discarded despite bidirectional formulation

Evaluation scope

- Only linear probing assessed — no fine-tuning evaluation
- No semantic structure captured for imagined speech at this scale

Conclusion

Contributions

- **EEG-RQVAE**: first VAE-based EEG encoder using residual vector quantization — produces discrete, hierarchical token sequences without masked autoencoding
- **2-stage pipeline**: Stage 1 decouples signal fidelity from Stage 2 semantic structure, enabling each objective to specialize
- **Fair benchmark**: unified preprocessing + linear probing on 12 BCI datasets under identical conditions

Key findings

- RVQ-Mamba is **best or second-best** on every evaluated dataset — discrete quantization does not sacrifice discriminability
- **Architecture alone is not the bottleneck**: baselines with similar parameters yield comparable results — the *training paradigm* matters more than the encoder design
- Stage 2 (JEPA) is the **only model** to rise above random k-NN consistency, confirming that latent prediction induces semantic structure that reconstruction cannot
- VAE-based approach opens new directions: CVAEs, discrete generative models

Future Work

Short term

- **Imagined speech**: focused evaluation on BCI 2020 T3, KaraOne, Chisco — phoneme vs. word-level decoding for speech-restoration BCIs
- **Fully bidirectional Mamba**: replace causal scan with true bidirectional backbone to match EEG's offline, non-causal nature
- **Loss design**: systematic exploration of Stage 1 weighting and Stage 1 / Stage 2 budget balance

Longer term

- **Scaling**: multi-corpus pretraining to characterize how representation quality scales with data volume and site diversity
- **Discrete generative modeling**: couple frozen RVQ tokenizer with an autoregressive Transformer over the code sequence — EEG generation, imputation, forecasting
- **CVAEs**: learn disentangled latent representations (e.g. task vs. subject identity)
- **Subject-specific models**: fine-tuned or distilled encoders for imagined speech, where inter-subject variability is extreme